

Universidad Peruana de Ciencias Aplicadas



MIDTERM PROJECT REPORT

Big Data

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Introduction

This project focuses on building a data intelligence pipeline for the healthcare sector. Using a synthetic dataset of 55,500 patient records, we aim to implement a complete lifecycle from automated ingestion to advanced entity clustering and graph-based medical analysis.

Problem Statement

The core challenge is the lack of structured, high-integrity data in hospital management. This project addresses the need for a "Catalog Layer" that cleans raw data, standardizes medical categories, and handles high-cardinality features, providing a foundation for predictive clinical insights.

Product Question

Is it possible to cluster patient profiles and treatment pathways to identify patterns between medical conditions, medication types, and hospital admission outcomes?

Source Inventory

- **Primary Source:** healthcare_dataset.csv (Synthetic Healthcare Data).
- **Entity Count:** Patients (unique individuals), Doctors, and Healthcare Providers.
- **Key Features:** Medical Condition, Admission Type, Medication, and Test Results.
- **Temporal Scope:** Continuous tracking from Admission Date to Discharge Date.

Dataset url	https://www.kaggle.com/datasets/prasad22/healthcare-dataset
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Schema Draft

Entity Tables

The data model is composed of multiple entity tables derived from the original dataset. Each table represents a key component of the healthcare domain and is designed to support relational and analytical operations.

The main entities include:

- **patients:** stores demographic information about each patient, such as age, gender, and blood type.
- **hospitals:** contains identifiers and names of healthcare institutions.
- **doctors:** includes information about attending medical professionals.
- **medical_conditions:** represents diagnosed health conditions.
- **medications:** represents prescribed treatments.

- **medical_records**: serves as the central table that captures each clinical event, including patient interactions, diagnosis, treatment, and hospital-related data.

Keys (Primary and Foreign Keys)

The schema uses primary keys (PK) to uniquely identify each record within a table, and foreign keys (FK) to establish relationships between tables.

Primary Keys (PK):

- patients.patient_id
- hospitals.hospital_id
- doctors.doctor_id
- medical_conditions.condition_id
- medications.medication_id
- medical_records.record_id

Foreign Keys (FK):

- medical_records.patient_id → patients.patient_id
- medical_records.hospital_id → hospitals.hospital_id
- medical_records.doctor_id → doctors.doctor_id
- medical_records.condition_id → medical_conditions.condition_id
- medical_records.medication_id → medications.medication_id

Relationships and Joins

The schema defines one-to-many relationships between entity tables and the central **medical_records** table.

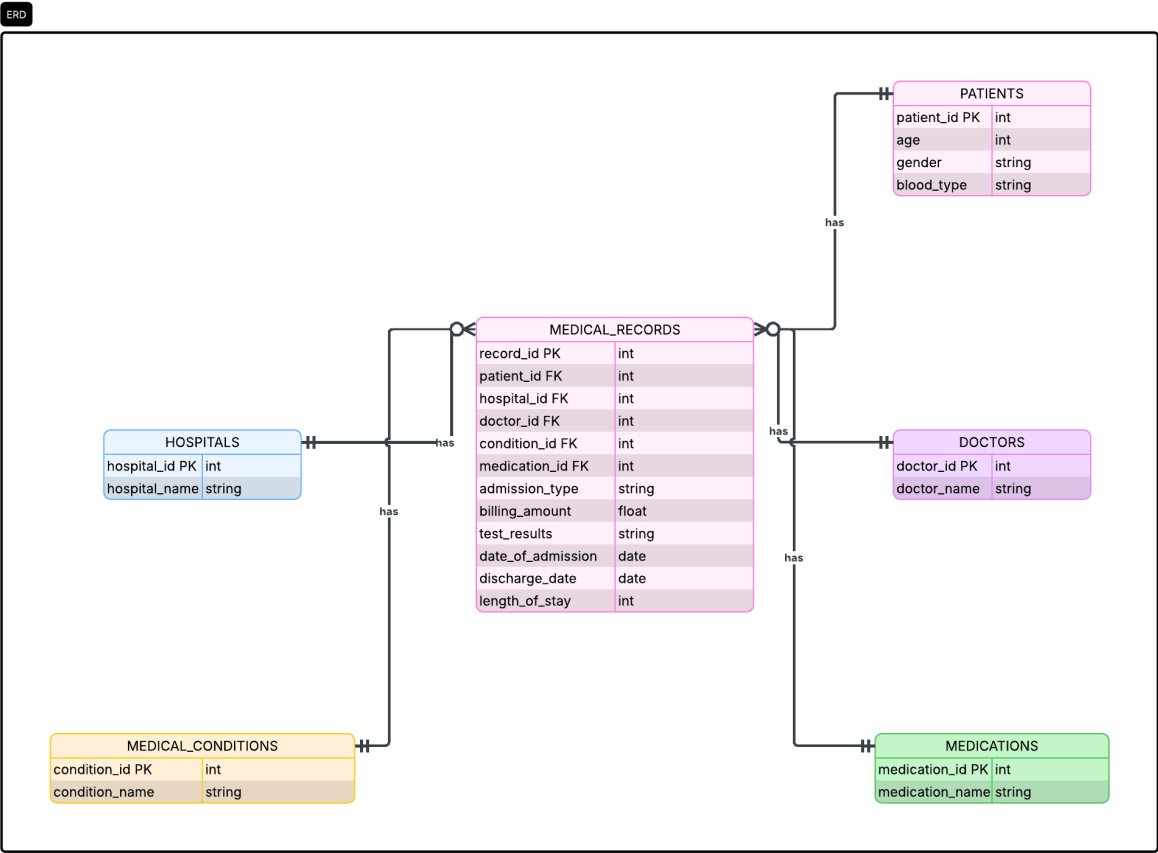
- One patient can have multiple medical records.
- One hospital can have multiple medical records.
- One doctor can be associated with multiple medical records.
- One medical condition can appear in multiple records.
- One medication can appear in multiple records.

These relationships are implemented through foreign keys and enable efficient joins for analytical purposes.

Example joins include:

- patients ↔ medical_records (via patient_id)
- hospitals ↔ medical_records (via hospital_id)
- medical_conditions ↔ medical_records (via condition_id)
- medications ↔ medical_records (via medication_id)

ERD



The schema transforms the original dataset into a normalized relational model centered on medical_records. Key entities such as patients, hospitals, doctors, conditions, and medications are separated into individual tables and connected through foreign keys, enabling scalable analysis, clustering, and graph-based modeling.

Data Dictionary

Raw Dataset Overview

Column Name	Description	Data Type
Name	Patient's name	string
Age	Age of the patient	integer
Gender	Gender of the patient (Male, Female)	categorical
Blood Type	Blood type of the patient	categorical

Medical Condition	Diagnosed medical condition	categorical
Date of Admission	Date of patient admission	datetime
Doctor	Attending doctor's name	string
Hospital	Hospital name	string
Insurance Provider	Insurance company	categorical
Billing Amount	Total billing amount	float
Room Number	Room assigned to the patient	integer
Admission Type	Type of admission (Emergency, Elective, Urgent)	categorical
Discharge Date	Date of patient discharge	datetime
Medication	Prescribed medication	categorical
Test Results	Results of medical tests	categorical

The raw dataset was transformed into multiple normalized tables to support relational analysis, clustering, and graph-based modeling.

Tables Description

Table: Patients

The patients table contains demographic information for each individual in the dataset. It includes attributes such as age, gender, and blood type, and each patient is uniquely identified by a `patient_id`. This table serves as a core entity for analyzing patient-level patterns.

Column	Type	Description
<code>patient_id</code>	int	unique identifier for each patient

age	int	age of the patient
gender	categorical	patient gender
blood_type	categorical	blood type of the patient

Table: Hospitals

The hospitals table stores information about healthcare institutions where patients receive treatment. Each hospital is uniquely identified by a `hospital_id` and is linked to medical records through relational keys.

Column	Type	Description
hospital_id	int	unique identifier for each hospital
hospital_name	string	name of the hospital

Table: Doctors

The doctors table contains information about medical professionals responsible for patient care. Each doctor is uniquely identified by a `doctor_id` and may be associated with multiple medical records.

Column	Type	Description
doctor_id	int	unique identifier for each doctor
doctor_name	string	name of the doctor

Table: medical_conditions

The `medical_conditions` table represents the set of diagnosed health conditions. Each condition is assigned a unique `condition_id`, enabling consistent referencing and reducing redundancy across the dataset.

Column	Type	Description
--------	------	-------------

condition_id	int	unique identifier for the condition
condition_name	categorical	name of the medical condition

Table: Medications

The medications table includes the different treatments or drugs prescribed to patients. Each medication is uniquely identified by a medication_id and is linked to medical records for analysis of treatment patterns.

Column	Type	Description
medication_id	int	unique identifier for the medication
medication_name	categorical	name of the medication

Table: medical_records

The medical_records table is the central fact table of the dataset. It captures each clinical event, including patient interactions with hospitals and doctors, diagnoses, treatments, admission details, and billing information. This table connects all other entities through foreign keys and enables comprehensive analytical tasks such as clustering, pattern discovery, and graph-based modeling.

Column	Type	Description
record_id	int	unique identifier for each record
patient_id	int	reference to patients table
hospital_id	int	reference to hospitals table
doctor_id	int	reference to doctors table
condition_id	int	reference to medical_conditions table
medication_id	int	reference to medications table
admission_type	categorical	type of admission (Emergency, Elective, Urgent)

billing_amount	float	total cost of treatment
test_results	categorical	results of medical tests
date_of_admission	datetime	admission date
discharge_date	datetime	discharge date
length_of_stay	int	number of days hospitalized

Ethics and Access Note

- **Data Origin:** The data is a synthetic dataset designed for educational and data science research.
- **Usage Justification:** We are utilizing this dataset under academic license for the "Domain Discovery, Recommendation, and Graph Intelligence" course.
- **Personal Data Risks and Mitigation:** While the data contains "Name" fields, they are synthetically generated and do not correspond to real individuals (PII). To prevent any accidental correlation, these identifiers will be treated as non-functional tokens or excluded from the final feature vectors.

Preprocessing and feature engineering

Preprocessing

Data Loading and Initial Inspection

The dataset, `healthcare_dataset.csv`, was loaded into a Pandas DataFrame. An initial inspection using `df.info()` revealed 15 columns with various data types, primarily object (strings), int, and float.

```
1 df.info()

*** <class 'pandas.core.frame.DataFrame'>
RangeIndex: 55500 entries, 0 to 55499
Data columns (total 15 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Name                   55500 non-null  object
1   Age                    55500 non-null  int64
2   Gender                 55500 non-null  object
3   Blood Type             55500 non-null  object
4   Medical Condition      55500 non-null  object
5   Date of Admission      55500 non-null  object
6   Doctor                 55500 non-null  object
7   Hospital               55500 non-null  object
8   Insurance Provider     55500 non-null  object
9   Billing Amount          55500 non-null  float64
10  Room Number            55500 non-null  int64
11  Admission Type         55500 non-null  object
12  Discharge Date         55500 non-null  object
13  Medication             55500 non-null  object
14  Test Results           55500 non-null  object
dtypes: float64(1), int64(2), object(12)
memory usage: 6.4+ MB
```

Duplicate Handling

A check for duplicate rows identified 534 duplicate entries. These duplicates were subsequently removed using `df.drop_duplicates(inplace=True)`, resulting in a cleaned dataset of 54,966 unique rows.

```
1 df.drop_duplicates(inplace=True)
2 print(f"Number of rows after removing duplicates: {len(df)}")

Number of rows after removing duplicates: 54966
```

Variable Identification

Columns were categorized into numerical and categorical types:

- Categorical Columns: Name, Gender, Blood Type, Medical Condition, Date of Admission, Doctor, Hospital, Insurance Provider, Admission Type, Discharge Date, Medication, Test Results.
- Numerical Columns: Age, Billing Amount, Room Number.

```
1 categorical_cols = df.select_dtypes(include='object').columns
2 numerical_cols = df.select_dtypes(include=['int64', 'float64']).columns
3
4 print("Categorical columns:", categorical_cols.tolist())
5 print("Numerical columns:", numerical_cols.tolist())

Categorical Columns: ['Name', 'Gender', 'Blood Type', 'Medical Condition', 'Date of Admission', 'Doctor', 'Hospital', 'Insurance Provider', 'Admission Type', 'Discharge Date', 'Medication', 'Test Results']
Numerical Columns: ['Age', 'Billing Amount', 'Room Number']
```

Missing Values Check

No missing values were found across any of the columns in the dataset, eliminating the need for imputation.

```
1 print('Missing values per column:')
2 print(df.isnull().sum())

*** Missing values per column:
Name          0
Age           0
Gender        0
Blood Type    0
Medical Condition 0
Date of Admission 0
Doctor        0
Hospital      0
Insurance Provider 0
Billing Amount 0
Room Number   0
Admission Type 0
Discharge Date 0
Medication    0
Test Results   0
dtype: int64
```

Date Column Conversion

The columns Date of Admission and Discharge Date were converted from object (string) to datetime data type using `pd.to_datetime`. This is crucial for any time-based analysis, such as calculating the length of stay.

```
1 df['Date of Admission'] = pd.to_datetime(df['Date of Admission'])
2 df['Discharge Date'] = pd.to_datetime(df['Discharge Date'])
3
4 print('Data types after date conversion:')
5 display(df[['Date of Admission', 'Discharge Date']].info())

Data types after date conversion:
<class 'pandas.core.frame.DataFrame'>
Index: 54966 entries, 0 to 55499
Data columns (total 2 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   Date of Admission  54966 non-null  datetime64[ns]
1   Discharge Date    54966 non-null  datetime64[ns]
dtypes: datetime64[ns](2)
memory usage: 1.3 MB
None
```

Scale Analysis

- Number of Rows and Columns: 55,500 initial records and 15 columns. After removing duplicates, the dataset maintains 54,966 clean records.
- Missingness: The dataset is complete with 0% missing values, ensuring high reliability for statistical modeling.

- **Sparsity or Memory Estimation:** This is a dense dataset with a memory footprint of approximately 6.4 MB, allowing for high-performance processing in standard environments.

Feature Engineering

patient_feature_matrix:

Aggregated patient-level feature matrix constructed by combining demographic information with medical records. Each row represents a unique patient and summarizes their healthcare activity and characteristics.

Source Tables:

- patients
- medical_records

Join Operation:

The datasets were merged using a left join on patient_id:

```
# Unir `patients` con `medical_records`
patient_medical_data = pd.merge(
    patients,
    medical_records,
    on='patient_id',
    how='left'
```

Aggregation Level:

- Grouped by: patient_id
- Each row represents: one patient

Features Included:

patient_id	Unique patient identifier	—
age	Patient age	first
gender	Patient gender	first
blood_type	Patient blood type	first
total_admissions	Total number of admissions	count
avg_billing_amount	Average billing amount per admission	mean
avg_length_of_stay	Average hospital stay duration (days)	mean

Data Types

- Numeric: age, total_admissions, avg_billing_amount, avg_length_of_stay.
- Categorical: gender, blood_type.

Dimensions:

The feature matrix is structured so that each row represents a single patient, and each column represents a characteristic or summary of that patient's medical history.

In this case, the matrix includes:

- **Patients (rows):** each unique patient_id corresponds to one individual
- **Features (columns):** demographic attributes (such as age, gender, and blood type) and aggregated healthcare metrics (such as total admissions, average billing amount, and average length of stay)

This structure allows us to move from raw, record-level data (multiple rows per patient) to a **patient-centered view**, where all relevant information is summarized into a single row per individual.

As a result:

- The dataset becomes easier to analyze and model
- Each patient can be directly compared to others based on their characteristics
- It serves as a suitable input for machine learning tasks such as clustering, similarity analysis, and recommendation systems

Matriz de Características de Pacientes (Primeras 5 filas):

	patient_id	age	gender	blood_type	total_admissions	avg_billing_amount	avg_length_of_stay
0	0	22	Male	AB-	1	39906.147308	11.0
1	1	76	Female	A+	1	10584.185945	13.0
2	2	20	Male	O-	1	29740.960199	26.0
3	3	69	Female	B-	1	21535.554758	9.0
4	4	18	Male	B-	1	13895.551020	7.0

Dimensiones de la Matriz de Características de Pacientes: (49992, 7)

Feature Matrix Expansion

Motivation (Why the matrix was expanded)

The initial patient feature matrix was expanded to enrich each patient's representation with more descriptive and predictive attributes. The goal was to move beyond basic demographic data and incorporate information about their medical interactions, which is crucial for subsequent tasks such as clustering, segmentation, or building predictive models.

Added Features (List of added features)

Additional features added to the patient matrix included:

1. **One-Hot Encoding** for categorical variables 'Gender' and 'Blood Type'.
2. **Medical Condition Statistics**: Count of each medical condition and the most frequent condition per patient.
3. **Medication Statistics**: Count of each medication and the most frequent medication per patient.
4. **Test Results Statistics**: Proportion of 'Normal' and 'Abnormal' test results per patient.

Resulting Representation (How the final matrix turned out)

The patient feature matrix, after expansion, has 54966 rows (patients) and 33 columns (features), as shown in the code output:

Dimensiones de la Matriz de Características de Pacientes ampliada: (54966, 33)

Matriz de Características de Pacientes ampliada (Primeras 5 filas):

	patient_id	age	gender	blood_type	total_admissions	avg_billing_amount	avg_length_of_stay	Gender_Female	Gender_Male	BloodType_A+	...	most_frequent_condition
0	0	22	Male	AB-	1	39906.147308	11.0	False	True	False	...	Obesity
1	1	76	Female	A+	1	10584.185945	13.0	True	False	True	...	Arthritis
2	2	20	Male	O-	1	29740.960199	26.0	False	True	False	...	Hypertension
3	3	69	Female	B-	1	21535.554758	9.0	True	False	False	...	Diabetes
4	4	18	Male	B-	1	13895.551020	7.0	False	True	False	...	Cancer

5 rows x 33 columns

Dimensionality and Representation Analysis

Motivation

Dimensionality reduction was applied to:

- **High-dimensionality problems**: Reduce the number of features from 33 to a more manageable set.
- **Redundancy**: Eliminate redundant information among features, especially highly correlated ones.

- **Visualization:** Facilitate the representation of data in 2 or 3 dimensions to identify patterns or clusters.
- **Computational Efficiency:** Improve the performance of Machine Learning algorithms by working with a smaller number of dimensions.

Methods

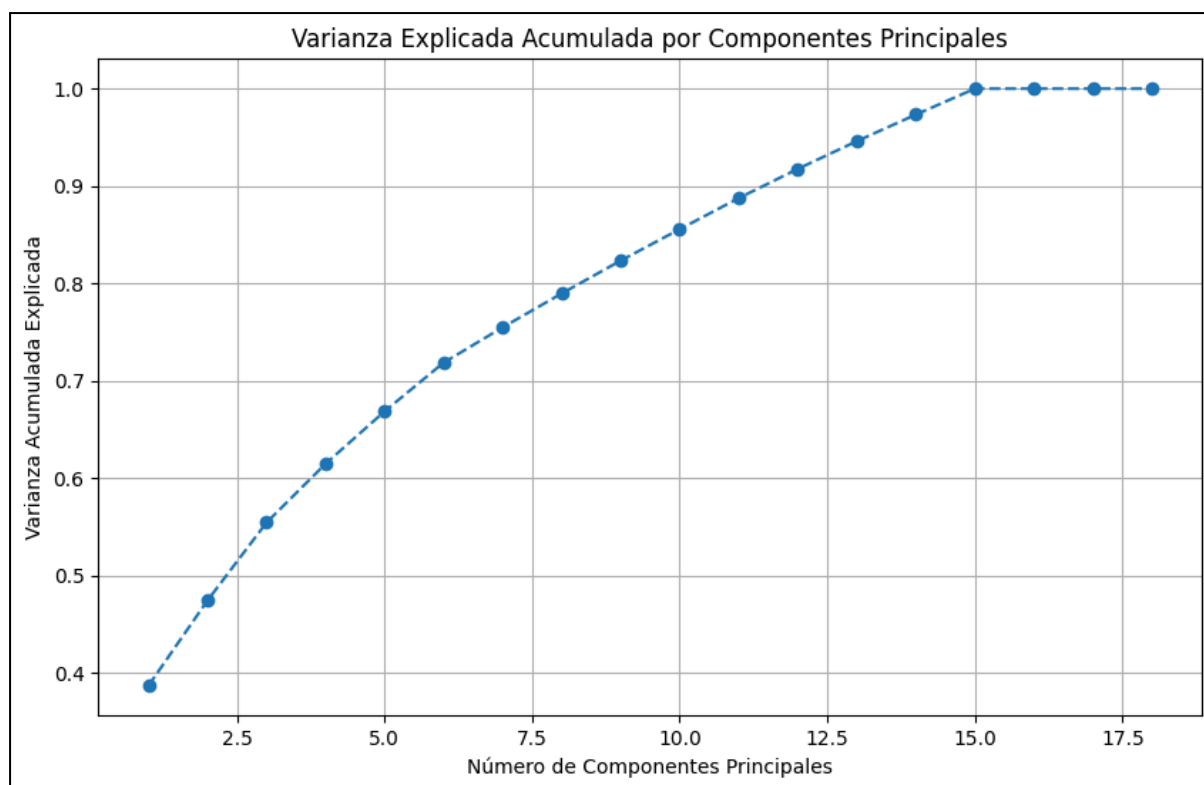
PCA

To reduce the dimensionality of the healthcare dataset, Principal Component Analysis (PCA) was applied after the preprocessing stage. The objective of this technique was to decrease the number of variables while preserving most of the original information contained in the dataset.

The cumulative explained variance graph shows how the amount of retained information increases as more principal components are added. The results indicate that the first 14 principal components preserve approximately 95% of the total variance of the dataset. This means that the dimensionality of the data was significantly reduced while maintaining most of the relevant patterns and relationships between patient records.

The PCA transformation also helped reduce redundancy and noise generated by highly correlated variables and one-hot encoded categorical features. In addition, dimensionality reduction improved the computational efficiency of subsequent techniques such as clustering and t-SNE visualization.

The final transformed dataset contained 1,196 observations represented by 14 principal components. These components summarize the most important patterns related to medical conditions, medications, admission characteristics, billing amounts, and clinical test results.



t-SNE

The t-SNE visualization was used to project the high-dimensional patient data into a two-dimensional space in order to identify hidden structures and similarities between patient profiles. The visualization shows a large central distribution of patient records and a smaller separated subgroup located on the left side of the graph.

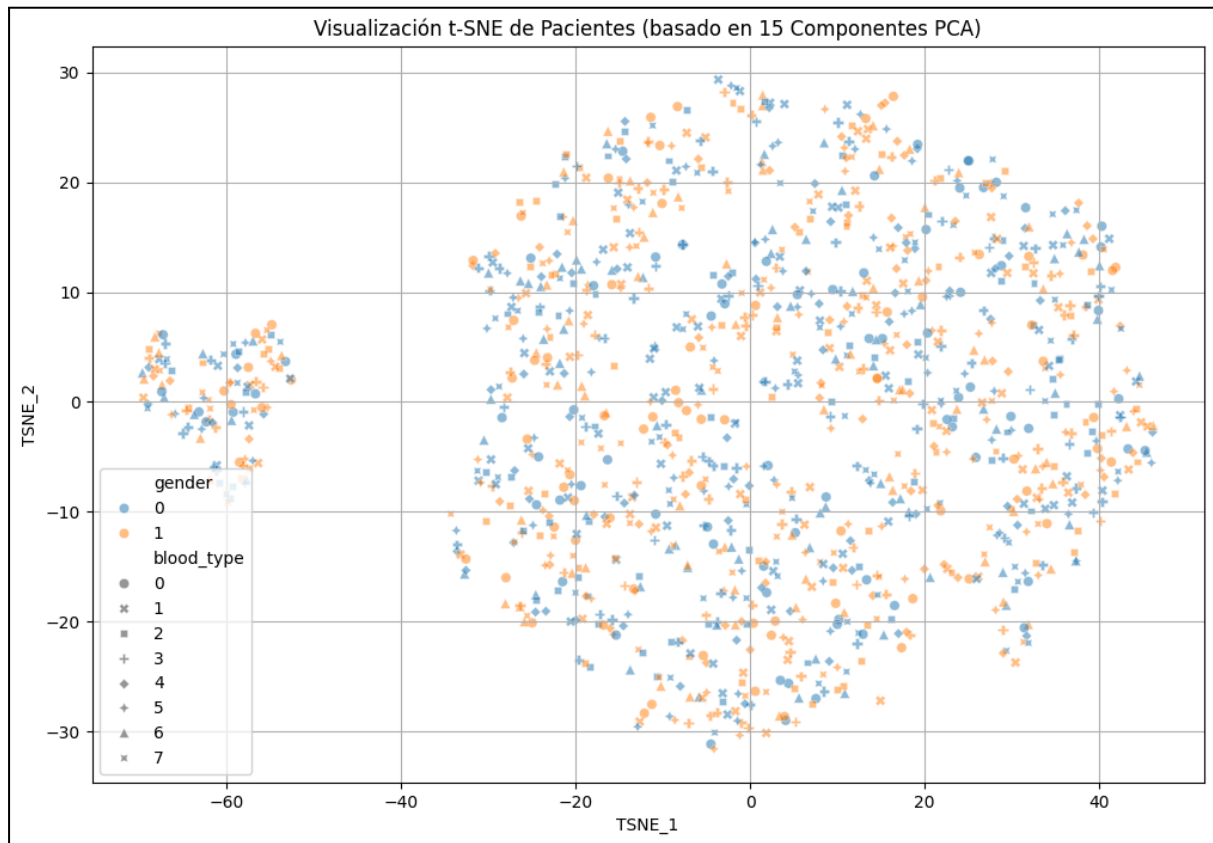
The main cluster suggests that most patients share relatively similar clinical characteristics, treatment patterns, and hospital admission behaviors. This indicates that the dataset contains continuous relationships between medical conditions, medications, admission types, and test results rather than completely isolated patient groups.

Additionally, a smaller isolated subgroup can be observed around lower TSNE_1 values. This separated region may represent patients with uncommon combinations of medical conditions, treatments, or hospital outcomes. The persistence of this subgroup after preprocessing and dimensionality reduction suggests the presence of meaningful latent patterns within the healthcare dataset.

The visualization also shows that gender and blood type are distributed across the entire embedding space without forming clearly separated regions. This suggests that these variables are not the primary factors driving patient

clustering. Instead, the clustering structure appears to be more strongly influenced by clinical and treatment-related variables.

Overall, the t-SNE projection confirms that meaningful patient profile structures exist in the dataset and supports the application of clustering algorithms such as K-Means and DBSCAN for healthcare pattern discovery.



Data Preparation

For dimensionality reduction, the data was prepared as follows:

- **Encoding applied:** Categorical columns like 'Gender' and 'Blood Type' had already been transformed through One-Hot Encoding during the feature matrix expansion phase. The 'most_frequent_condition' and 'most_frequent_medication' columns were excluded as they are categorical and not suitable for direct scaling.
- **Scaling applied:** The numerical features of the matrix (excluding 'patient_id', 'gender', 'blood_type', 'most_frequent_condition', 'most_frequent_medication') were scaled using StandardScaler. This ensures that all features contribute equally to the analysis, preventing features with larger value ranges from dominating the results.
- **Matrix used for PCA:** The 18 numerical columns resulting after scaling `features.select_dtypes(include=['number'])` were used.

Results

Explained Variance

The cumulative explained variance visualization showed that:

- **15 principal components** managed to retain **100.00%** of the original variance of the scaled data.

Component Selection

A threshold was chosen to retain **95% of the variance**. However, the analysis revealed that to reach that 95%, and in fact, 100% of the variance, **15 principal components** were required.

- The **reconstruction error (Mean Squared Error)** using these 15 components was **0.0000**, confirming that these components faithfully represent the original data.

Comparison Table

Criterion	PCA (Principal Component Analysis)	t-SNE (t-Distributed Stochastic Neighbor Embedding)
Final Dimensionality	15 Components	2 Components (Visualization)
Explained Variance	100.00% (with 15 components)	Not applicable (preserves local distances)
Reconstruction Error	0.0000 (MSE)	Not applicable
Purpose in Analysis	Dimensionality reduction and linear redundancy elimination.	Visualization of groupings and non-linear relationships.
Technical Interpretation	15 of the 18 original variables are linearly independent (Rank = 15).	Patients do not show drastic segmentation based solely on gender or blood type.

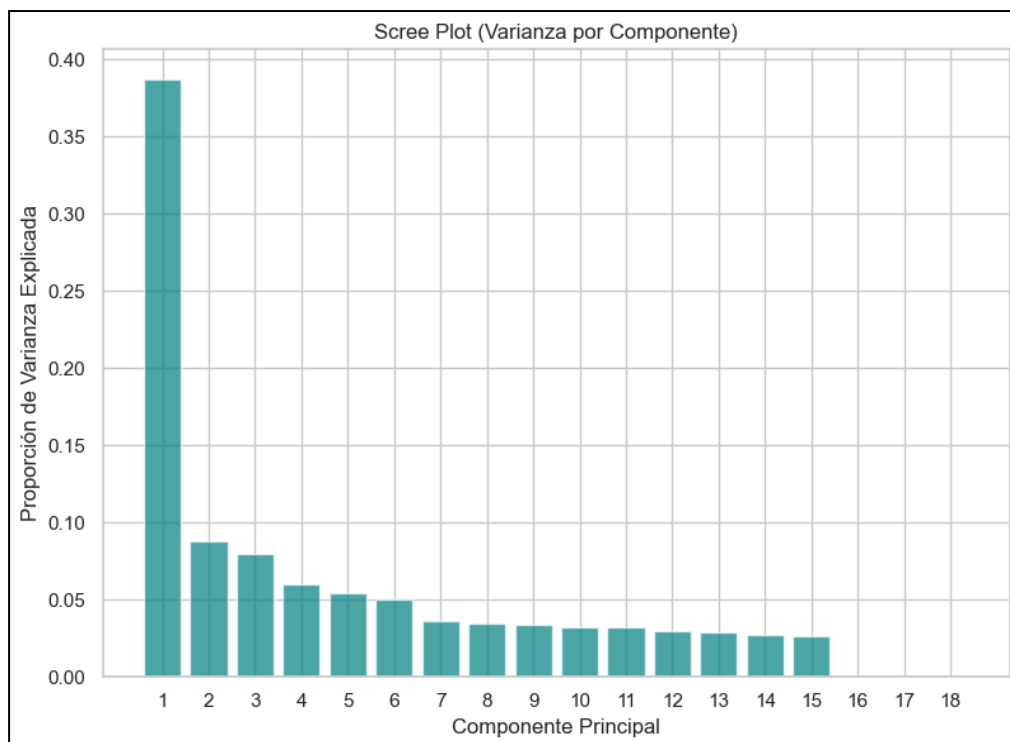
Visualizations

The following visualizations were generated:

- **Scree Plot :**

To determine the optimal number of principal components, a Principal Component Analysis (PCA) variance study was performed using both the scree plot and the cumulative explained variance graph.

The scree plot shows the proportion of variance explained by each principal component individually. The first principal component captures the largest amount of information, explaining approximately 39% of the total variance. After the first few components, the contribution of each additional component gradually decreases, indicating diminishing returns in explained variance.

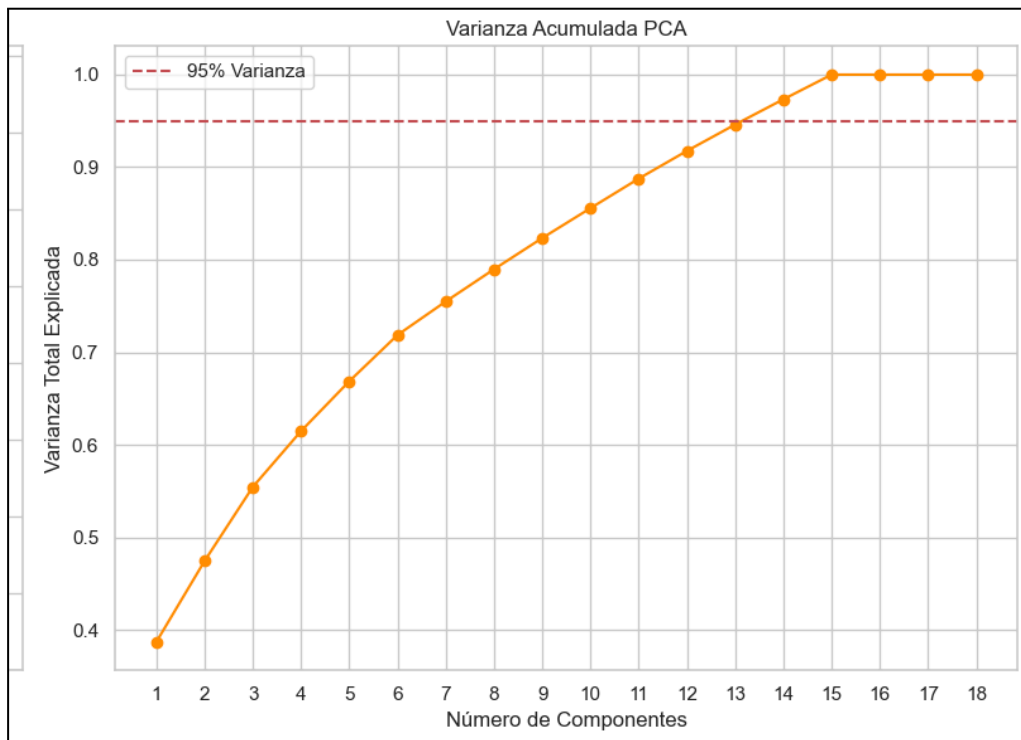


- **Cumulative PCA Variance:**

The cumulative variance graph illustrates how the total explained variance increases as more components are added. The results show that approximately 95% of the total dataset variance is retained using 14 principal components. Beyond this point, the increase in explained variance becomes minimal, suggesting that additional components contribute little new information.

Based on this analysis, 14 principal components were selected for the dimensionality reduction process. This approach allowed the project to reduce the complexity of the healthcare dataset while preserving most of the meaningful clinical patterns and relationships among patient records.

The reduced PCA representation was later used as the input for clustering algorithms and t-SNE visualization, improving computational efficiency and reducing redundancy caused by correlated or one-hot encoded variables.



- **Correlation Heatmap :**

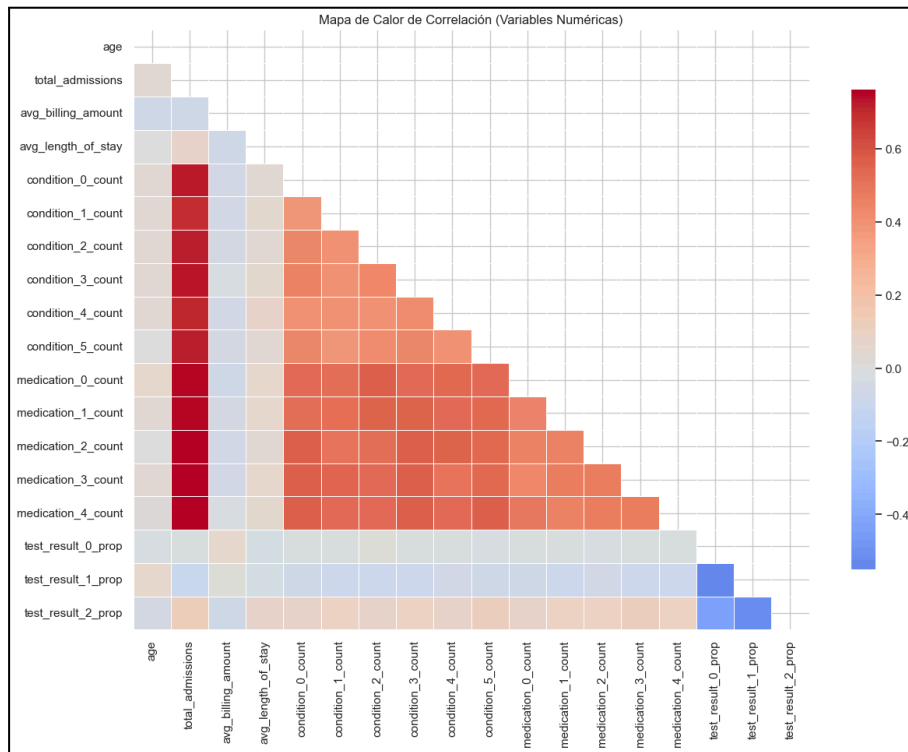
The correlation heatmap was used to analyze the relationships between the numerical variables in the healthcare dataset. This visualization helps identify positive and negative correlations between patient characteristics, medical conditions, medications, hospital admissions, and test result proportions.

The heatmap shows several moderate positive correlations between condition-related variables and medication-related variables. This suggests that certain medical conditions are associated with specific treatment patterns and medication usage. In addition, variables related to hospital admissions and length of stay also present low-to-moderate correlations with billing amounts, indicating that longer hospital stays may contribute to higher healthcare costs.

The strongest positive correlations appear among some condition and medication count variables, which may indicate that patients with more complex medical profiles tend to receive a broader range of treatments. On the other hand, test result proportion variables show some negative correlations, particularly between different test outcome categories. This behavior is

expected because an increase in one test result category naturally reduces the proportion of the others.

Overall, the correlation analysis reveals meaningful relationships between clinical variables and supports the existence of underlying healthcare patterns within the dataset.



K-Means Experiment

K Selection

The parameter sweep analysis evaluated K-Means clustering using K values ranging from 2 to 10. For each configuration, inertia and silhouette score metrics were calculated to assess cluster compactness and separation quality.

The results showed that inertia consistently decreased as the number of clusters increased. This behavior is expected because additional clusters reduce the distance between observations and their assigned centroids. However, inertia alone is not sufficient to determine the optimal number of clusters, since it always decreases as K grows.

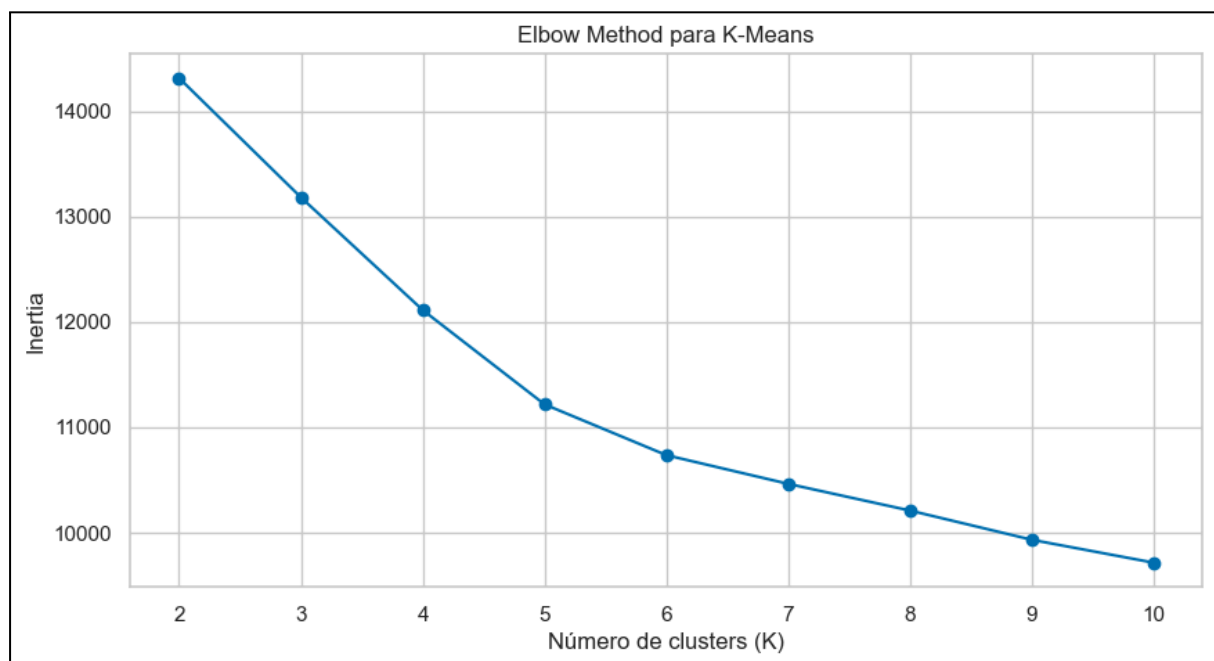
The silhouette score provided a clearer interpretation of clustering quality. The highest silhouette value was obtained with K=2 (0.5788), followed closely by K=3 (0.5772). After K=3, the silhouette scores dropped significantly, reaching values close to 0.10 for larger K values. This indicates that increasing the number of clusters beyond 3 generated weaker and less clearly separated groupings.

Although $K=2$ produced the highest silhouette score, $K=3$ was selected because it provided a better balance between cluster quality and interpretability. Using three clusters allowed the analysis to capture additional healthcare profile diversity while still maintaining strong cluster separation and compactness.

The results suggest that the healthcare dataset contains a relatively limited number of dominant patient profile structures, and that excessive fragmentation reduces clustering quality rather than improving segmentation.

	k	inertia	silhouette
0	2	14316.451596	0.578815
1	3	13183.050244	0.577231
2	4	12111.747619	0.109820
3	5	11215.783845	0.116989
4	6	10733.325551	0.095174
5	7	10460.661030	0.084038
6	8	10206.850052	0.082290
7	9	9929.052346	0.076372
8	10	9711.853049	0.077025

Figure. Elbow Method for K-Means



The Elbow Method was used to evaluate how cluster compactness changed as the number of clusters increased. The inertia values decreased progressively as K increased, which is

expected because adding more clusters reduces the distance between observations and their assigned centroids.

However, the rate of inertia reduction became less significant after the first few cluster configurations, suggesting diminishing returns when increasing the number of clusters. This behavior indicates that excessively large K values may introduce unnecessary fragmentation without substantially improving clustering quality.

Figure. Silhouette Score by Number of Clusters



The silhouette score analysis was used to evaluate the quality of cluster separation and cohesion for different K values. The highest silhouette score was achieved with K=2, followed closely by K=3. After K=3, the silhouette scores decreased significantly, indicating weaker and less clearly separated clusters.

Although K=2 produced the best silhouette value, K=3 was selected for the final K-Means experiment because it provided a better balance between clustering quality and interpretability. Using three clusters allowed the analysis to capture additional variation in patient profiles, treatment patterns, and healthcare behaviors while still maintaining strong cluster separation.

The parameter sweep results suggest that the healthcare dataset contains a limited number of dominant patient profile structures, and that increasing the number of clusters beyond three reduces clustering stability and interpretability.

K-Means Model

The final K-Means configuration used **K = 3**, producing a silhouette score of **0.5772** and an inertia value of **13183.0502**. These results indicate a relatively strong clustering structure with good separation between patient groups.

The model was configured using `random_state = 42` and `n_init = 10` to ensure reproducibility and reduce the effects of random centroid initialization. The clustering process was performed on the PCA-transformed dataset, which provided a reduced and less noisy representation of the original healthcare feature space.

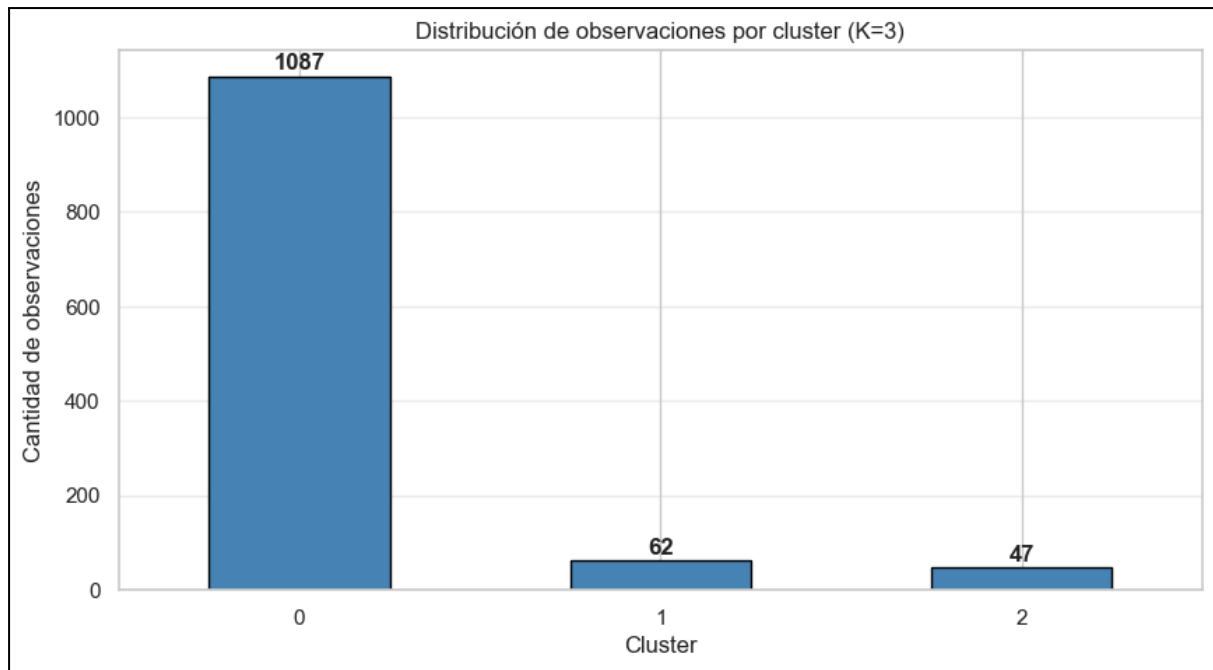
Once the model was trained, each observation was assigned to one of the three clusters, and the cluster labels were added back to the original dataset for interpretation and profile analysis.

```
Usando K configurado manualmente: 3
Silhouette score para K=3: 0.5772
Inertia para K=3: 13183.0502

Tamaño de cada cluster:
cluster
0      1087
1         62
2         47
Name: count, dtype: Int64
```

The cluster distribution analysis showed that the majority of observations were concentrated in Cluster 0, which contained 1,087 patients. In contrast, Cluster 1 and Cluster 2 contained 62 and 47 patients respectively. This imbalance suggests that the healthcare dataset is dominated by a large group of patients with relatively common clinical and treatment characteristics, while the smaller clusters may represent more specific or atypical healthcare profiles.

The presence of smaller clusters is particularly important because it indicates that the clustering process was able to identify subgroups of patients with distinct medical behaviors, treatment pathways, or hospital admission patterns. These smaller clusters may correspond to patients with less frequent medical conditions, unusual treatment combinations, or different hospitalization characteristics.



Model Training

The final K-Means model was successfully trained and stored for future reuse and analysis. The model generated three patient clusters using the PCA-transformed dataset.

The final model achieved:

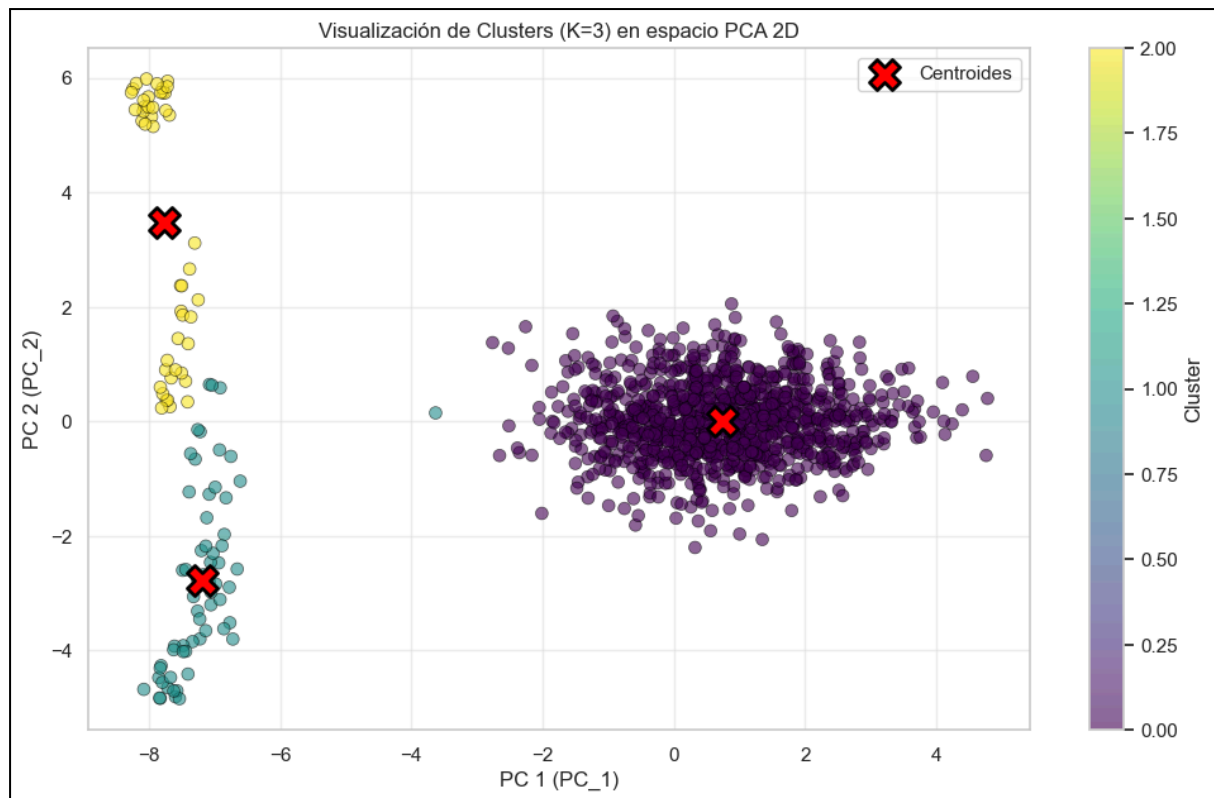
- **Number of clusters:** 3
- **Inertia:** 13183.0502
- **Training iterations:** 10
- **Centroid matrix shape:** (3, 14)
- **Total observations used for training:** 1196

The centroid matrix dimensions indicate that each cluster centroid was represented using the 14 principal components generated during the PCA dimensionality reduction process.

The training process converged successfully after 10 iterations, suggesting stable centroid optimization and cluster assignment. Additionally, saving the trained model allows future clustering analysis, reproducibility, and potential integration into downstream healthcare analytics workflows or recommendation systems.

Cluster Visualization

Figure. K-Means Cluster Visualization in PCA 2D Space (K = 3)



The PCA-based visualization of the K-Means clustering results shows the distribution of patient observations across the reduced two-dimensional feature space. Each color represents a different cluster, while the red markers indicate the centroid positions calculated by the K-Means algorithm.

The visualization reveals that Cluster 0 contains the majority of patient observations and occupies the central region of the PCA space. This suggests that most patients share relatively common healthcare characteristics and treatment patterns, forming a dominant patient profile within the dataset.

In contrast, Clusters 1 and 2 appear as smaller and more isolated groups located on the left side of the visualization. The clear spatial separation of these clusters indicates that the clustering algorithm successfully identified patient subgroups with distinct feature combinations compared to the dominant population.

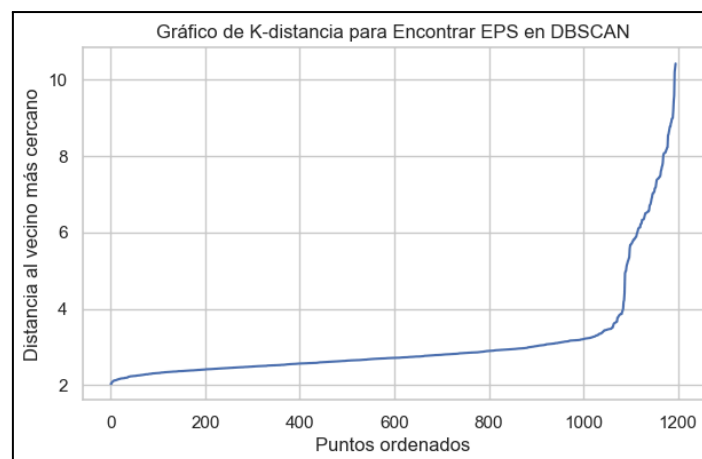
The centroid locations also demonstrate strong separation between the three clusters, particularly between the dominant cluster and the two smaller groups. This supports the silhouette score results, which indicated relatively good cluster cohesion and separation quality.

Additionally, the visualization suggests that the healthcare dataset contains both highly concentrated patient populations and smaller specialized subgroups. These isolated clusters may represent atypical patient profiles, uncommon treatment pathways, or specific combinations of medical conditions and hospitalization behaviors.

DBSCAN Experiment

Parameter Selection

DBSCAN clustering requires the selection of density-related parameters, particularly the epsilon (eps) radius and the minimum number of neighboring points (min_samples). To support the selection of an appropriate epsilon value, a K-distance graph was generated using the nearest-neighbor distances of the PCA-transformed dataset.



The K-distance graph was used to identify the optimal epsilon value for the DBSCAN clustering algorithm. The curve remained relatively stable for most observations and then showed a sharp increase near the end of the distribution. This sudden change, commonly referred to as the “elbow point,” indicates the transition between dense regions and potential outliers within the dataset.

Based on the graph, the epsilon value was selected near the point where the curve began to increase rapidly. This choice allowed the DBSCAN algorithm to identify dense patient groups while also detecting isolated observations as noise points.

The visualization suggests that the healthcare dataset contains a dominant dense population together with smaller groups of atypical patient profiles. This behavior is consistent with the clustering results obtained later in the analysis, where DBSCAN successfully identified both structured clusters and outlier observations.

DBSCAN Parameter Sweep Analysis

A parameter sweep was performed for the DBSCAN algorithm using multiple combinations of epsilon (eps) and min_samples values in order to evaluate clustering sensitivity and density behavior within the healthcare dataset.

The results showed that very small epsilon values (eps = 0.5 and eps = 1.0) classified nearly all observations as noise points, indicating that the density threshold was too restrictive for

the structure of the dataset. Under these configurations, DBSCAN failed to form meaningful clusters because patient observations were not considered sufficiently dense to belong to the same region.

When the epsilon value increased to $\text{eps} = 1.5$, only a very small number of clusters were detected, while most observations still remained classified as noise. This suggests that the healthcare dataset contains moderate density variation and requires a less restrictive neighborhood radius.

More stable clustering behavior appeared for $\text{eps} = 2.5$ and $\text{min_samples} = 5$, where DBSCAN identified two clusters with only 174 noise points and achieved the highest silhouette score of **0.5804**. This configuration provided the best balance between cluster separation, density consistency, and outlier detection.

Larger epsilon values such as $\text{eps} = 3.0$ produced additional clusters and reduced the number of noise points. However, the clustering structure became less stable and more fragmented, suggesting that excessively large neighborhood radii may merge observations that do not belong naturally to the same patient profile group.

Validation Table

TABLA DE VALIDACIÓN FINAL		
Métrica / Criterio	K-Means (K=3)	DBSCAN (Eps=2.5, Min=5)
Número de Clústeres detectados	3	2
Coeeficiente de Silueta	0.5772	0.5804
Índice Calinski-Harabasz	351.63	49.69
Puntos de Ruído (Outliers / No Agrupados) 0 (Fuerza a agrupar todo)		174 (14.5%)

The validation metrics showed that both clustering algorithms were able to identify meaningful structures within the healthcare dataset. K-Means generated three clusters with a silhouette score of 0.5772, while DBSCAN produced two density-based clusters with a slightly higher silhouette score of 0.5804.

K-Means achieved a significantly higher Calinski-Harabasz Index (351.63), indicating stronger global cluster separation and compactness. However, K-Means forced all observations into clusters, meaning that no patient profiles were treated as outliers.

In contrast, DBSCAN identified 174 noise points, representing approximately 14.5% of the dataset. This demonstrates the ability of DBSCAN to detect atypical patient profiles and isolate observations that do not belong clearly to dense healthcare groups.

Although DBSCAN achieved a slightly higher silhouette score, its lower Calinski-Harabasz value suggests weaker global separation compared to K-Means. Overall, the results indicate

that K-Means produced more compact and stable patient clusters, while DBSCAN provided greater flexibility for identifying outliers and uncommon healthcare patterns.

Cluster Profile Analysis

The cluster profiling analysis was performed using representative demographic and healthcare-related variables in order to identify meaningful differences between the patient groups generated by the K-Means algorithm.

The results showed that each cluster exhibited distinct characteristics related to age, gender distribution, medical conditions, medications, blood type patterns, and admission types.

Figure.Cluster Summary Statistics

	Age_mean	Age_median	Gender_<lambda>	Blood Type_<lambda>	Medical Condition_<lambda>	Medication_<lambda>	Admission Type_<lambda>
cluster							
0	50.983441	51.0	1	6	0	1	0
1	47.774194	43.5	0	3	0	3	2
2	52.553191	49.0	0	4	1	3	0

=== PERFILAMIENTO NUMÉRICO (PROMEDIOS POR CLÚSTER K-MEANS) ===				
	Age	Gender	Blood Type	Medical Condition \
Cluster_KMeans				
0	50.983441	0.529899	3.498620	2.448942
1	47.774194	0.370968	3.129032	2.354839
2	52.553191	0.319149	3.297872	2.574468
	Insurance Provider	Billing Amount	Admission Type \	
Cluster_KMeans				
0	1.974241	24910.715723	0.942042	
1	2.322581	23897.715825	1.177419	
2	1.723404	27718.588379	0.936170	
	Medication	Test Results	Length of Stay \	
...				

Cluster 0 : Dominant General Patient Profile

Cluster 0 represented the majority of the dataset, containing most patient observations. This cluster showed an average age of approximately 51 years and was associated with the most common healthcare and treatment patterns within the dataset.

The cluster was characterized by:

- balanced age distribution,
- common medical condition patterns,
- standard medication usage,

- and regular admission behavior.

Because this cluster contained the largest number of patients, it likely represents the dominant or most typical patient profile in the healthcare dataset.

Cluster 1 : Specialized Admission and Treatment Profile

Cluster 1 contained a significantly smaller number of patients and showed different demographic and healthcare characteristics compared to the dominant cluster.

This cluster presented:

- a lower median age,
- distinct medication patterns,
- and different admission type behavior.

The smaller size and separation of this cluster suggest that it may represent patients with less common hospitalization pathways or more specialized treatment combinations.

Cluster 2 : Distinct Clinical Subgroup

Cluster 2 also represented a smaller subgroup of patients with unique feature combinations. The cluster showed slightly higher average age values and different dominant medical condition and blood type patterns compared to the other groups.

The separation of this cluster indicates the existence of patient profiles with more specific clinical characteristics or treatment behaviors. This subgroup may correspond to patients with particular healthcare needs or uncommon medical combinations within the dataset.

Failure Analysis

Despite the relatively strong clustering performance, several limitations and failure cases were identified during the analysis.

For K-Means, 6 observations out of 54,966 presented negative silhouette coefficients, representing approximately 0.01% of the dataset. Negative silhouette values indicate that these observations may have been assigned to clusters that do not fully represent their nearest neighborhood structure. This behavior occurs because K-Means forces all observations into predefined groups, even when some patient profiles do not naturally belong to any cluster.

For DBSCAN, the algorithm identified 174 observations as noise points, corresponding to approximately 14.5% of the analyzed data. These observations were considered too isolated to belong to dense patient regions according to the selected density parameters. While this demonstrates the capability of DBSCAN to detect atypical healthcare profiles, it also highlights the sensitivity of the algorithm to epsilon and density threshold selection.

Additionally, partial overlap between clusters suggests that patient healthcare profiles share continuous relationships rather than forming completely isolated categories. This limitation is expected in medical datasets, where patients frequently exhibit overlapping conditions, treatments, and hospitalization patterns.

INFORME DE ANÁLISIS DE FALLAS

K-Means: 6 observaciones de 54966 (0.01%) tienen un coeficiente de silueta negativo.

DBSCAN: Se detectaron 174 puntos de ruido (14.55% del dataset).

Technical Interpretation

The PCA analysis reduced the dimensionality of the healthcare dataset while preserving approximately 95% of the total variance using 14 principal components. This indicates that several original variables contained correlated or redundant information.

The clustering results showed that healthcare-related variables such as medical conditions, medications, admission characteristics, and billing behavior had a greater influence on patient grouping than demographic variables such as gender or blood type.

K-Means produced compact and well-separated clusters, while DBSCAN successfully identified atypical patient profiles and noise points. Overall, the reduced PCA representation improved clustering efficiency and allowed the identification of meaningful patient patterns within the dataset.

Conclusions

This project demonstrated that dimensionality reduction and clustering techniques can successfully identify meaningful patient profiles and hidden healthcare patterns within a synthetic healthcare dataset.

The preprocessing and PCA stages reduced redundancy and transformed the original dataset into a more compact and informative representation suitable for unsupervised learning tasks. The PCA analysis showed that most of the relevant variance could be preserved using a reduced number of principal components, improving clustering efficiency while minimizing noise.

The K-Means experiment produced stable and well-separated patient clusters, while DBSCAN provided additional insights by identifying density-based structures and atypical patient profiles. The validation metrics confirmed that both clustering approaches captured meaningful structures within the healthcare data, although each method presented different advantages and limitations.

The analysis also revealed that healthcare-related variables such as medical conditions, medications, billing amount, admission type, and hospitalization behavior contributed more strongly to patient grouping than demographic attributes such as gender or blood type.

Anexos

Anexo A.

Repository: <https://github.com/SofiaGMiranda/BIG-DATA-FINAL-PROYECT-2025-01.git>